THE PROBLEM

Factories have data, not decisions.

Sensors stream all day, yet failures still arrive as surprises.

Alarms fire too late

Fixed thresholds trip when the machine is already close to failure.

Knowledge is buried

Technicians dig through PDF manuals and old work orders by hand.

Downtime is expensive

Unplanned stops cost thousands of dollars every hour.

THE USER

Maintenance engineers and shift supervisors.

WHAT THEY NEED

- Know **early** that a machine is drifting
- Know **why**, with evidence they can check
- Know **what to do**, safely

WHAT THEY HAVE TODAY

- Threshold alarms and spreadsheets
- PDF manuals and scattered work orders
- Tribal knowledge in a few senior heads

THE SOLUTION

Maintenance Copilot: catch failures early.

01 Drift detection

Each machine is compared to its own baseline, not a fixed limit.

02 Grounded diagnosis

Cause, confidence, urgency, action and safety note, with cited sources.

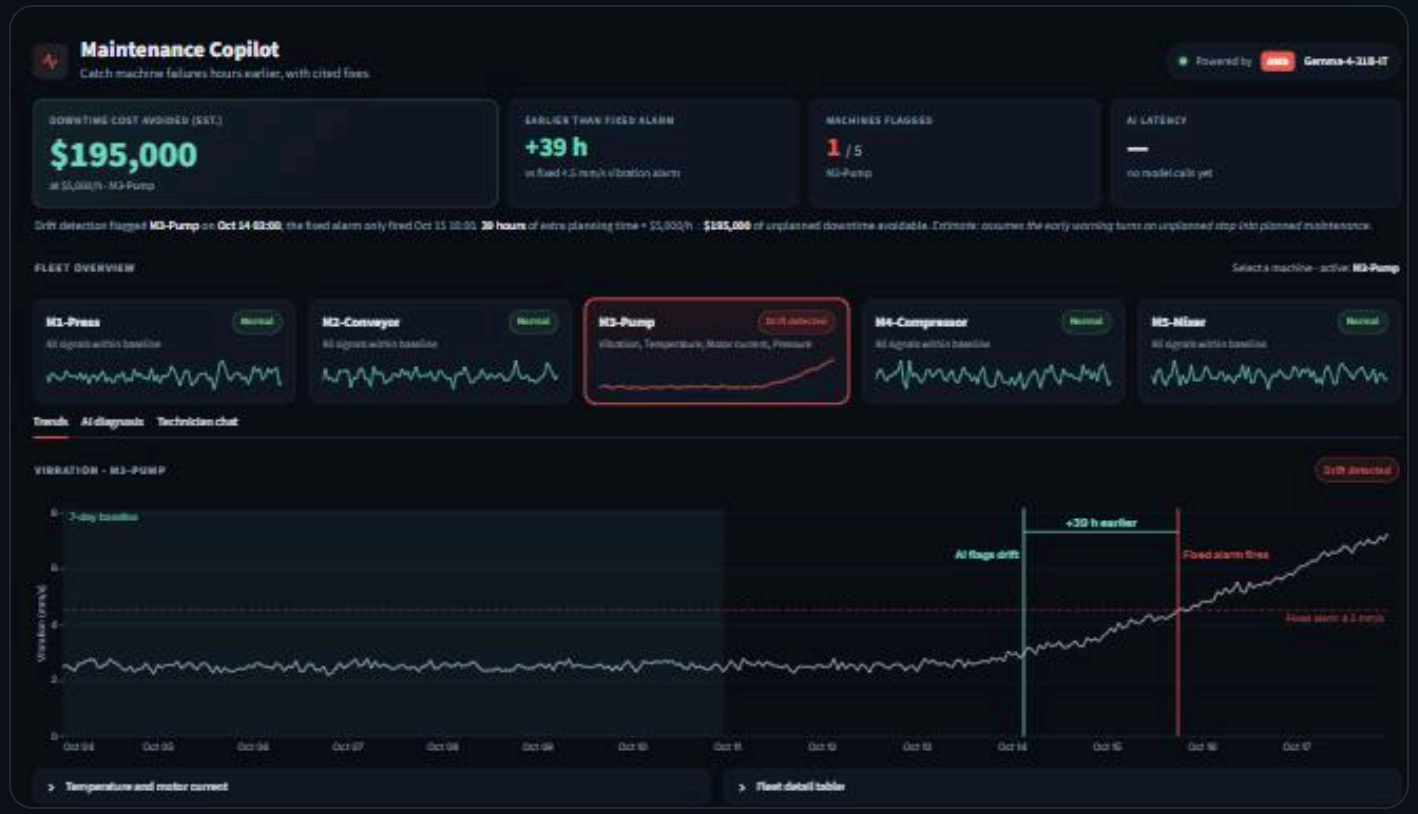
03 Technician chat

Ask questions answered only from the manual and maintenance log.

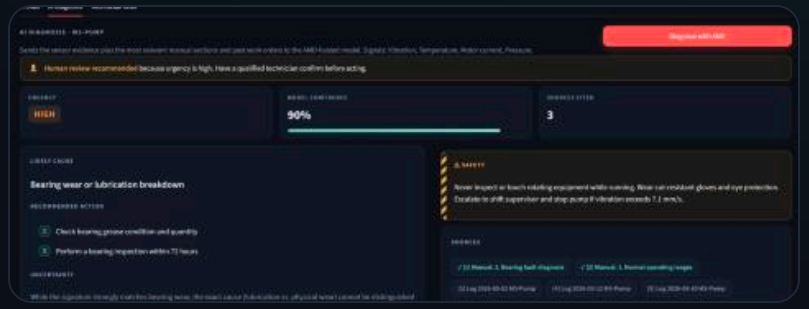
04 Human review flag

Raised when the model is unsure or urgency is high.

LIVE DEMO · M3-PUMP BEARING FAULT



1 · Detect. M3-Pump flagged red. The AI flags drift **39 h** before the fixed 4.5 mm/s alarm fires.



2 · Diagnose. One click: cited cause, urgency, confidence and a safety callout.



3 · Ask. “How do I replace the drive-end bearing safely?” with source chips.

HOW AMD POWERS IT

All inference runs on AMD Token Factory.

Gemma-4-31B-IT through an OpenAI-compatible API, returning structured JSON.

INPUT
Sensor readings

→

DETECT
Drift vs. machine baseline

→

RETRIEVE
Manual + log (TF-IDF top-k)

→

AMD TOKEN FACTORY
Gemma-4-31B-IT

→

GUARD
Validate JSON, safe fallback

DIAGNOSIS

≈ 6–9 s ~1,000 tokens

CHAT ANSWER

≈ 7 s ~870 tokens

LIVE AI ENGINE PANEL

Endpoint, model, latency per call and tokens used, shown in the app.

MEASURABLE IMPACT

Hours of warning, not minutes.

EARLIER THAN FIXED ALARM

+39 h

vs. a fixed 4.5 mm/s vibration alarm (demo data)

DOWNTIME AVOIDED (EST.)

$195k

≈ $195,000 at $5,000/h, configurable in the app

TIME TO ANSWER

Seconds

cited, safety-aware actions instead of manual search

Estimate assumes the early warning turns an unplanned stop into planned maintenance

NEXT STEPS

From demo to the plant floor.

DATA

Live sensor streams

MQTT / OPC-UA

MODELS

Embeddings + vision

Defect photos, on AMD

WORKFLOW

CMMS work orders

Create tickets directly

VALIDATION

Plant pilot

Real production data

Maintenance Copilot

Detect early · diagnose with sources · act safely

AMD Token Factory · Gemma-4-31B-IT

github.com/Vaish54224/maintenance-copilot